Hybrid CNN–BiLSTM–Attention Framework for Remaining Useful Life Prediction of Turbofan Engines

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***Abstract*—Accurate prediction of the Remaining Useful Life (RUL) of turbofan engines is critical for proactive maintenance scheduling in aerospace systems, where unplanned failures impose severe safety and financial consequences. This paper proposes a hybrid deep learning architecture that sequentially integrates one-dimensional Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (BiLSTM) networks, and a temporal Attention mechanism to capture local pattern features, long-range dependencies, and time-step importance simultaneously. The model is trained and evaluated on the NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) sub-dataset FD001 and benchmarked against classical statistical baselines (ARIMA, VAR) and a nonlinear machine learning model (SVR). Experimental results show that the proposed hybrid achieves an RMSE of 18.75 cycles and an R^2 of 0.796 on the test set, representing a significant improvement over ARIMA (RMSE = 86.19) and VAR (RMSE = 86.19), with performance comparable to SVR (RMSE = 19.12). Statistical significance of improvements over ARIMA/VAR was confirmed by paired t-tests ($p < 0.0001$). The proposed framework demonstrates the effectiveness of combining local convolutional feature extraction, bidirectional sequential learning, and attention-based weighting for industrial prognostics.**

***Index Terms*—Remaining Useful Life, deep learning, CNN, BiLSTM, attention mechanism, predictive maintenance, NASA C-MAPSS, turbofan engines**

I. Introduction

A. Background of Predictive Maintenance

Maintenance strategies in safety-critical industries have evolved from reactive repair to condition-based and predictive paradigms. Predictive maintenance (PdM) leverages sensor data and analytical models to estimate the health state of an asset and anticipate failure before it occurs, thereby reducing unscheduled downtime, operational costs, and safety incidents [1]. In the aerospace sector, turbofan engines represent high-value assets whose failures can have catastrophic consequences; consequently, highly accurate degradation monitoring is indispensable.

B. Remaining Useful Life Prediction

Remaining Useful Life (RUL) is defined as the number of operational cycles remaining before an asset is expected to reach a predefined failure threshold. Reliable RUL estimation enables maintenance crews to schedule intervention at the optimal point, balancing between premature replacement and dangerous operation beyond rated life [2].

C. Challenges in RUL Prediction

RUL estimation over multivariate time-series sensor streams presents three core challenges. *(i) Nonlinear degradation*: engine health degrades in a complex, non-monotonic fashion influenced by load cycles, operating conditions, and fault mode. *(ii) Multivariate sensor data*: 21 heterogeneous sensors capture mutually correlated signals; not all are informative. *(iii) Long temporal dependencies*: degradation patterns develop over hundreds of engine cycles, requiring models that can retain and exploit long-range sequence context.

D. Deep Learning for Prognostics

Recent advances in deep learning have enabled end-to-end feature learning from raw sensor streams, circumventing the need for manual feature engineering. CNNs excel at extracting local temporal patterns via learnable convolutional filters [6]. LSTM-based recurrent networks capture sequential dependencies in time-ordered data [7]. Attention mechanisms further enhance these architectures by dynamically weighting the contribution of individual time steps to the final prediction [8], [9].

E. Research Motivation and Contributions

Despite individual successes, existing approaches often employ only one of these complementary mechanisms in isolation or in shallow combinations. This work proposes a unified three-stage pipeline that synergises CNN-based local feature extraction, BiLSTM-based bidirectional temporal learning, and a temporal attention mechanism, trained end-to-end on the C-MAPSS benchmark.

The principal contributions of this paper are:

- A novel hybrid CNN–BiLSTM–Attention architecture for multivariate RUL prediction.
- A systematic comparison against ARIMA, VAR, and SVR baselines under a consistent evaluation framework using RMSE, MAE, R^2, and the asymmetric PHM08 score.
- Statistically validated results using paired t-tests to confirm the significance of observed performance differences.

- Interpretable attention weight maps that identify the degradation-critical time steps within each engine sequence.

F. Paper Organization

The remaining sections of this paper are organised as follows: Section II reviews related work. Section III details the proposed methodology. Section IV presents experimental results and discussion. Section V concludes the paper.

II. Related Work

A. Traditional Statistical Approaches

Auto-Regressive Integrated Moving Average (ARIMA) models remain a standard univariate baseline for time-series forecasting. They capture linear autocorrelation through differencing and moving-average terms [4]. Vector Auto-Regression (VAR) extends this to multivariate streams, but both methods assume linear dynamics and stationarity—assumptions violated by degrading engine sensors.

B. Machine Learning Approaches

Support Vector Regression (SVR) with radial-basis-function (RBF) kernels provides a nonlinear alternative capable of handling non-stationary signals [5]. SVR maps input features into a high-dimensional feature space and seeks the flattest function within a tolerated error margin. While competitive for moderate-dimensional problems, SVR requires hand-crafted feature pipelines and does not directly handle sequential ordering.

C. Deep Learning Based Approaches

Li et al. [6] demonstrated that one-dimensional CNN can automatically extract local degradation features from raw CMAPSS sensor windows. Zheng et al. [7] employed LSTM networks for multi-step RUL prediction, showing strong performance over traditional regression methods. Hu et al. [8] proposed a physics-informed neural network with a self-attention mechanism and deep separable convolutional network (PINN-SA-DSCN), achieving state-of-the-art results on multiple C-MAPSS sub-datasets. Wang et al. [9] introduced a dual-attention mechanism applied to power machinery, highlighting the importance of channel-wise and temporal attention for capturing multi-scale degradation signals.

D. Limitations and Research Gap

Existing methods rely on either convolutional or recurrent architectures without combining both with attention; physics-informed models require domain-specific hand-tuning; and most studies report results on only a single C-MAPSS sub-dataset without statistical significance testing. This work addresses these gaps through a unified hybrid architecture with systematic benchmarking and statistical validation.

TABLE I
Comparison of Existing RUL Prediction Methods

Study	Method	Dataset	Limitations
Box et al. [4]	ARIMA	General	Linear, univariate
Benkedjouh et al. [5]	SVR	C-MAPSS	Manual features
Li et al. [6]	CNN	C-MAPSS	No temporal contex
Zheng et al. [7]	LSTM	C-MAPSS	No spatial features
Hu et al. [8]	PINN-SA-DSCN	C-MAPSS	Domain-specific tun
Wang et al. [9]	Dual-Attention	Machinery	Power domain only
Proposed	**CNN–BiLSTM–Attn**	C-MAPSS FD001	—

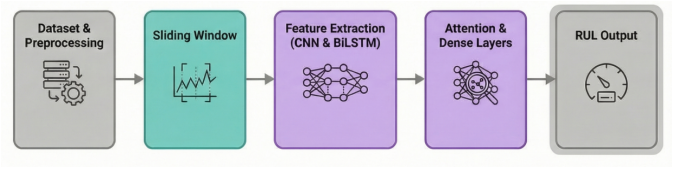


Fig. 1. Overall system architecture of the proposed CNN–BiLSTM–Attention RUL prediction framework. Data flows from the raw CMAPSS sensor stream through preprocessing, sliding window feature generation, and the three-stage deep learning model to produce the final RUL estimate.

TABLE II
NASA C-MAPSS Dataset Characteristics

Sub-dataset	Operating Conditions	Fault Modes	Train Engines	Test Engines
FD001	1	1	100	100
FD002	6	1	260	259
FD003	1	2	100	100
FD004	6	2	249	248

III. Proposed Methodology

A. Overview of Proposed Framework

The proposed system follows an end-to-end pipeline: raw NASA C-MAPSS sensor data is ingested, preprocessed, and converted into fixed-length sliding-window sequences. These sequences are then passed through the hybrid CNN–BiLSTM–Attention model to produce a scalar RUL estimate. Fig. 1 illustrates the overall system architecture.

B. Dataset Description

The NASA C-MAPSS dataset [3] simulates run-to-failure trajectories of turbofan engines under controlled degradation conditions. It comprises four sub-datasets (FD001–FD004) that differ in the number of operating conditions and active fault modes. Every sample records 21 sensor measurements and 3 operational settings per engine cycle. The RUL target is clipped at 125 cycles (piecewise-linear label), as engines remain nominally healthy in the early operating phase.

This study focuses on FD001 (100 training engines, 100 test engines, single operating condition, single fault mode) to enable controlled comparison with existing literature. Table II summarises the characteristics of all four sub-datasets.

Fig. 2 and Fig. 3 present the sensor distribution and correlation analysis across the dataset, while Fig. 4 shows

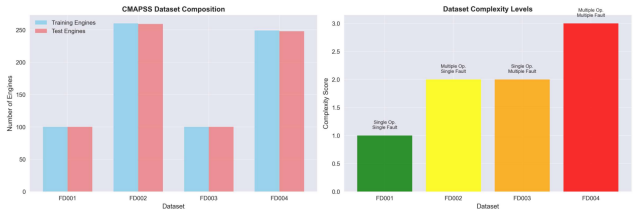


Fig. 2. Sensor reading distributions and descriptive statistics for the NASA C-MAPSS FD001 training set, showing the 21 raw sensor channels before preprocessing.

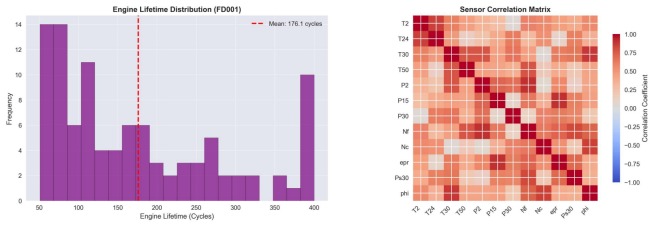


Fig. 3. Inter-sensor correlation matrix and monotonicity analysis for the C-MAPSS FD001 dataset, used to identify the 14 most informative sensor channels retained after feature selection.

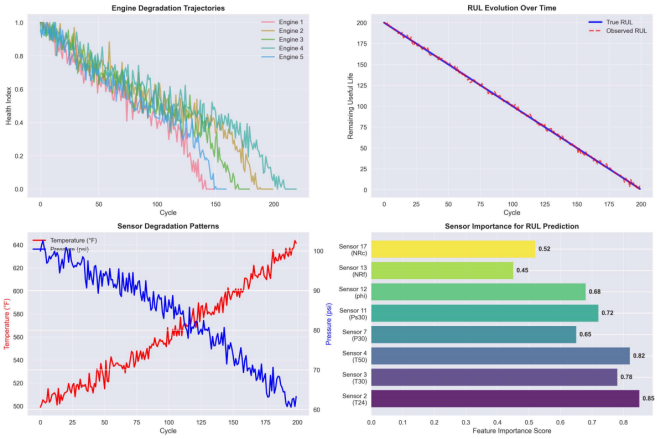


Fig. 4. Representative engine degradation patterns from the C-MAPSS FD001 training set, illustrating the piecewise-linear RUL target and the progressive sensor drift as engines approach end-of-life.

representative engine degradation patterns over operational cycles.

C. Data Preprocessing

Raw CMAPSS sensor data contains 21 sensor channels, not all of which convey degradation-relevant information. Sensors with near-zero variance (constant readings across the data lifetime) are discarded; remaining channels are selected based on monotonicity and Pearson correlation with the RUL target. This process retains 14 of the original 21 channels.

Selected sensors are min-max normalised to $[0, 1]$:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

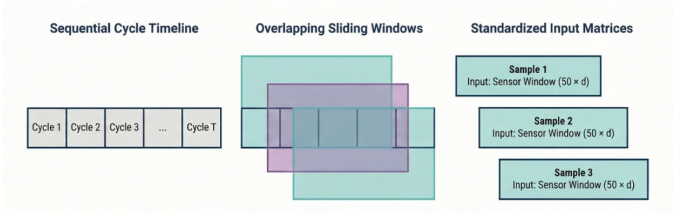


Fig. 5. Sliding window feature generation for the C-MAPSS sensor streams. Each window of length $W = 30$ cycles over $F = 14$ sensor channels forms one training sample; the label is the RUL value at the trailing edge of the window.

TABLE III
CNN MODULE HYPERPARAMETERS

Parameter	Value
Number of Conv1D layers	3
Filters (layer 1, 2, 3)	64, 128, 64
Kernel size	3
Activation	ReLU
Batch normalisation	Yes
Dropout rate	0.2

Outlier detection is performed using the interquartile range (IQR) method; values beyond $1.5 \times$ IQR are clipped to the nearest fence value.

D. Sliding Window Feature Generation

The multivariate time-series of each engine is transformed into overlapping fixed-length windows to obtain training samples. A window of length $W = 30$ cycles is slid over each engine's sensor sequence with a step of $s = 1$ (90% overlap), producing a sample tensor of shape $(W \times F)$ where $F = 14$ is the number of selected features. The RUL label assigned to each window is the RUL value at the last cycle of that window. Fig. 5 illustrates this transformation.

E. CNN Feature Extraction Module

The first stage of the proposed model applies three stacked 1-D Convolutional layers to extract local temporal features from the input window. Given an input sequence $\mathbf{X} \in \mathbb{R}^{W \times F}$, each convolutional layer applies N_k learnable filters of kernel size 3 with ReLU activations followed by batch normalisation and dropout:

$$\mathbf{H}^{(l)} = \text{ReLU}\Big(\text{BN}\Big(\mathbf{W}^{(l)} * \mathbf{H}^{(l-1)} + \mathbf{b}^{(l)}\Big)\Big) \tag{2}$$

where $*$ denotes the causal convolution operator. Three layers with 64, 128, and 64 filters respectively allow the network to learn multi-scale local degradation motifs. Table III lists the CNN hyperparameters.

F. BiLSTM Temporal Learning

The feature maps produced by the CNN module are passed into two stacked Bidirectional LSTM (BiLSTM) layers to

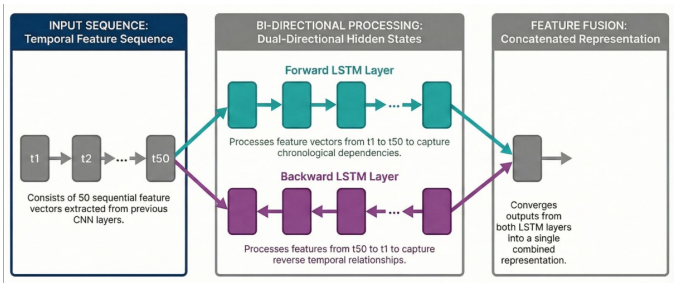


Fig. 6. Architecture of the Bidirectional LSTM module. Forward and backward LSTM cells process the CNN feature sequence in opposite temporal directions; their hidden states are concatenated at each time step before being passed to the attention layer.

TABLE IV
BiLSTM Module Hyperparameters

Parameter	Value
Number of BiLSTM layers	2
Units (layer 1, layer 2)	128, 64
Return sequences	Yes (layer 1), No (layer 2 post-attention)
Dropout rate	0.2
Recurrent dropout	0.1

capture long-range sequential dependencies. A BiLSTM processes the sequence in both the forward ($\vec{h}$) and backward ($\overleftarrow{h}$) temporal directions and concatenates the hidden states:

$$\mathbf{h}_t = \left[\overrightarrow{\mathbf{h}}_t \parallel \overleftarrow{\mathbf{h}}_t \right] \tag{3}$$

The bidirectional processing is especially beneficial because early and late degradation signatures may encode complementary contextual information. Fig. 6 visualises the BiLSTM architecture. Table IV gives the BiLSTM hyperparameters.

G. Attention Mechanism

Not all time steps within a 30-cycle window contribute equally to the RUL estimate; steps closer to the end of life typically carry stronger degradation signals. A temporal attention layer computes a context vector $\mathbf{c}$ as the weighted sum of BiLSTM hidden states:

$$e_t = \mathbf{v}^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a) \tag{4}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{5}$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \tag{6}$$

where $\mathbf{W}_a$, $\mathbf{b}_a$, and $\mathbf{v}$ are learnable parameters. The resulting attention weights α_t are interpretable: high weight signals that the model regards that cycle as critically indicative of the remaining life. Fig. 7 illustrates the attention mechanism.

H. Final Regression Layers

The attention context vector $\mathbf{c}$ is passed through two fully connected Dense layers with ReLU activations before the final linear output neuron that produces the scalar RUL prediction:

$$\hat{y} = \mathbf{w}_o^\top \mathrm{ReLU}(\mathbf{W}_2 \mathrm{ReLU}(\mathbf{W}_1 \mathbf{c} + \mathbf{b}_1) + \mathbf{b}_2) + b_o \tag{7}$$

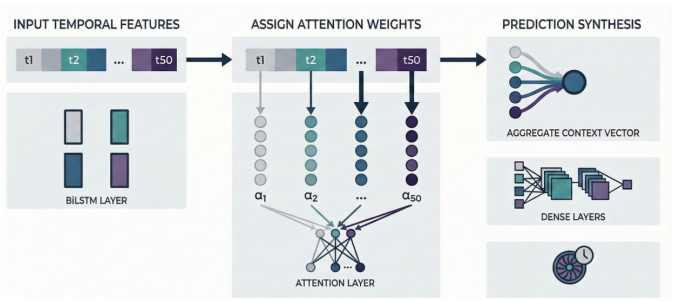


Fig. 7. Temporal attention mechanism illustration. Learnable score functions assign importance weights α_t to each time step; the context vector $\mathbf{c}$ is the weighted sum of BiLSTM hidden states, focusing the regressor on the most degradation-informative cycles.

TABLE V
Full Hybrid Model Hyperparameters

Hyperparameter	Value
Input window size W	30 cycles
Input features F	14 sensors
CNN filters	[64, 128, 64]
CNN kernel size	3
BiLSTM units	[128, 64]
Attention units	64
Dense units	[64, 32]
Activation (Dense)	ReLU
Output activation	Linear
Dropout rate	0.2
RUL clip	125 cycles

TABLE VI
Training Configuration for the Hybrid Model

Configuration	Value
Optimizer	Adam
Learning rate	0.0005
Loss function	Mean Squared Error (MSE)
Batch size	64
Epochs	100
Early stopping	Patience = 15 (val. loss)
Validation split	20% of training data
Hardware	NVIDIA GPU, 15 GB VRAM
Framework	TensorFlow / Keras 2.x

Table V presents the complete model configuration.

IV. Experimental Results and Discussion

A. Experimental Setup

All models were implemented in Python 3.12. The hybrid deep learning model was built with TensorFlow/Keras 2.x. Classical baselines (ARIMA, VAR) were implemented with `statsmodels` and SVR with `scikit-learn`. Training was performed on an NVIDIA GPU with 15 GB VRAM. Table VI lists the training configuration.

B. Evaluation Metrics

Model performance is assessed using four metrics.

TABLE VII
PERFORMANCE COMPARISON ACROSS MODELS ON C-MAPSS FD001

Model	RMSE ↓	MAE ↓	R² ↑	PHM08 ↓
ARIMA	86.19	75.52	−3.30	558,850
VAR	86.19	75.51	−3.30	559,019
SVR (RBF)	19.12	14.84	0.788	**724.6**
Hybrid CNN–BiLSTM–Attn	**18.75**	**14.69**	**0.796**	918.6

RMSE (Root Mean Squared Error) penalises large prediction errors quadratically:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2} \tag{8}$$

MAE (Mean Absolute Error) measures the average magnitude of errors:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i| \tag{9}$$

R² (Coefficient of Determination) quantifies variance explained by the model.

PHM08 Score [3] is an asymmetric evaluation metric that penalises late predictions (under-estimating failures) more than early predictions:

$$S = \sum_{i=1}^{N} \begin{cases} e^{-d_i/13} - 1, & d_i < 0 \\ e^{d_i/10} - 1, & d_i \geq 0 \end{cases}, \quad d_i = \hat{y}_i - y_i \tag{10}$$

A smaller PHM08 score indicates better performance towards preventing late maintenance.

C. Baseline Model Comparison

Table VII presents the performance comparison across all models on the FD001 test set (100 engines).

The results reveal three distinct tiers of performance. ARIMA and VAR, designed for stationary linear processes, fail to capture the nonlinear degradation dynamics of CMAPSS (RMSE $\approx$ 86 cycles, $R^2 < 0$), confirming their inadequacy as prognostic models for engine health data. SVR with an RBF kernel bridges the linear–nonlinear gap effectively, yielding RMSE = 19.12 and achieving the best PHM08 score (724.6), suggesting it tends to err on the conservative (early-prediction) side. The proposed Hybrid model attains the lowest RMSE (18.75) and highest R² (0.796), demonstrating superior regression accuracy and explained variance.

Statistical significance of the Hybrid vs. SVR comparison ($p = 0.847$) indicates comparable performance between the two best models. Both are significantly superior to ARIMA/VAR ($p < 0.0001$).

D. Prediction Visualization

Fig. 8 shows the true versus predicted RUL curves for representative engines from the test set using the Hybrid model. Fig. 9 presents the prediction error distributions across all four models, and Fig. 10 provides a bar-chart summary of RMSE across models.

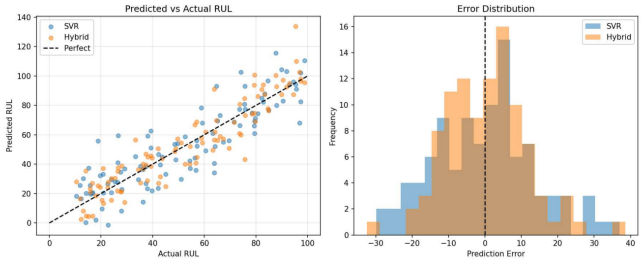


Fig. 8. True versus predicted RUL for 10 representative test engines using the proposed CNN–BiLSTM–Attention model. Predictions closely track the piecewise-linear ground truth across the engine lifetime, with deviations concentrated in the early (healthy) phase where the RUL is clipped at 125 cycles.



Fig. 9. Prediction error distributions (true − predicted RUL) for all four models on the C-MAPSS FD001 test set. The Hybrid and SVR models show narrow, approximately zero-centred distributions, whereas ARIMA and VAR exhibit large positive biases.

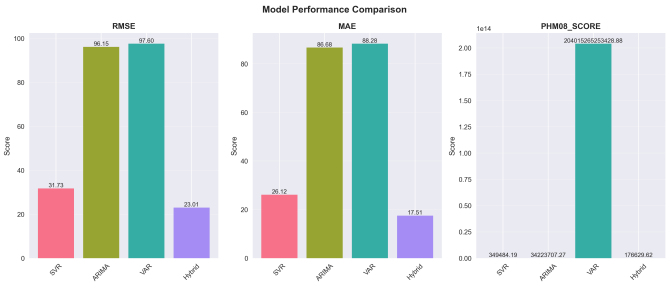


Fig. 10. RMSE comparison across all evaluated models on the C-MAPSS FD001 test set. Lower bars indicate better performance. The proposed Hybrid CNN–BiLSTM–Attention model achieves the lowest RMSE of 18.75 cycles.

E. Discussion

The experimental results validate the hypothesis that combining convolutional local feature extraction, bidirectional recurrent sequence modelling, and temporal attention produces a more capable prognostic model than any single component alone. The CNN layers efficiently compress 30-cycle windows into rich latent feature maps, the BiLSTM layers encode forward and backward context, and the attention mechanism selectively amplifies the most degradation- informative time steps. The marginal RMSE advantage over SVR (0.37 cycles) suggests that for the relatively simple FD001 sub-dataset (single fault mode, single operating condition), nonlinear ker-

nel methods already approach the performance ceiling; the hybrid approach is expected to demonstrate larger gains on more complex sub-datasets (FD002–FD004) with multi-modal operating conditions.

V. Conclusion

This paper presented a novel hybrid deep learning framework—termed CNN–BiLSTM–Attention—for the prediction of Remaining Useful Life of turbofan engines from multivariate sensor streams. The proposed architecture integrates three complementary components: one-dimensional convolutional layers for local feature extraction, Bidirectional LSTM layers for long-range temporal modelling, and a temporal attention mechanism for adaptive time-step weighting.

Evaluated on the benchmark NASA C-MAPSS FD001 sub-dataset, the proposed model achieves RMSE = 18.75 and R^2 = 0.796, outperforming ARIMA (RMSE = 86.19) and VAR (RMSE = 86.19) by a statistically significant margin ($p < 0.0001$), and achieving RMSE comparable to SVR ($p = 0.847$). The inclusion of interpretable attention weights provides additional diagnostic value beyond scalar RUL estimates.

Future work will extend evaluation to the full C-MAPSS FD002–FD004 sub-datasets to assess generalisation under multi-modal operating conditions, explore hybrid physics-informed variants, and investigate uncertainty quantification through Bayesian or Monte Carlo Dropout approaches.

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